

TITLE PAGE

- **Problem Statement ID – 26184**
- **Problem Statement Title –** Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID – 137425**
- **Team Name – Team NUEVION**



Detailed Solution

A complaint-to-interception platform that traces fraud money to its exit card, predicts the ATM, requests an officer-approved silent hold, and dispatches police

- **Voice Intake (~5 s):** Bhashini + Whisper convert 1930-style calls (simulated) into structured case JSON via rule-based field extraction (UTR, UPI ID, amount, fraud type).
- **Mule Trace (~300 ms):** Neo4j maps L1 → L2 → L3 hops and flags the terminal exit card.
- **ATM Forecast (~45 min):** Random Forest ranks ATMs near the criminal's last known call location (simulated cell-tower data), using highway proximity, crime density and withdrawal limits.
- **Silent Interdiction (~60 s):** Alert-free debit-lien on officer approval, plus auto-drafted §106 BNSS notice.
- **Command GIS:** Threat heatmaps with city, crime and time filters, and nearest-patrol routing.

We move police from the complaint desk to the cash-out point.

How ArthaVyūh Addresses the Problem

Existing Systems	ArthaVyūh
Freezes leftover funds; Layer-1 mule sharing	Complaint becomes a ~45-min ATM forecast
Officer gets no predicted cash withdrawal locations	Top-3 ATMs ranked near the criminal's last known call location
Manual tracing across mule layers and banks	Neo4j trace of 3 layers in ~300 ms to the exit account
Visible freezes can alert the account holder.	Silent, time-boxed lien, officer-approved, no alert
Split workflows: legal notice and dispatch handled separately	Auto-drafted §106 BNSS notice + GIS patrol routing in ~60 s

Innovation and Uniqueness of the Solution

Multilingual Voice Intake:

Bhashini ingestion with Whisper offline fallback converts voice complaints to structured case JSON.



Closed-Loop Interdiction:

One complaint drives the whole chain: trace, forecast, hold, dispatch.



Case-Specific ATM Ranking.:

Random Forest ranks the likely cash-out ATMs near the criminal's last known call location (simulated cell-tower data), not a city-wide hotspot list.

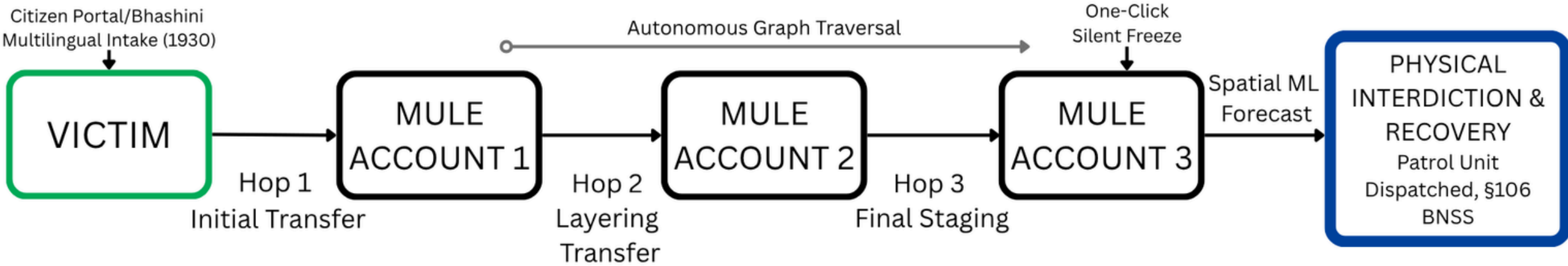


Officer-Approved Silent Hold:

Time-boxed debit-lien with an auto-drafted §106 BNSS notice



HOW ARTHAVYŪH WORKS



YOUTUBE VIDEO LINK:

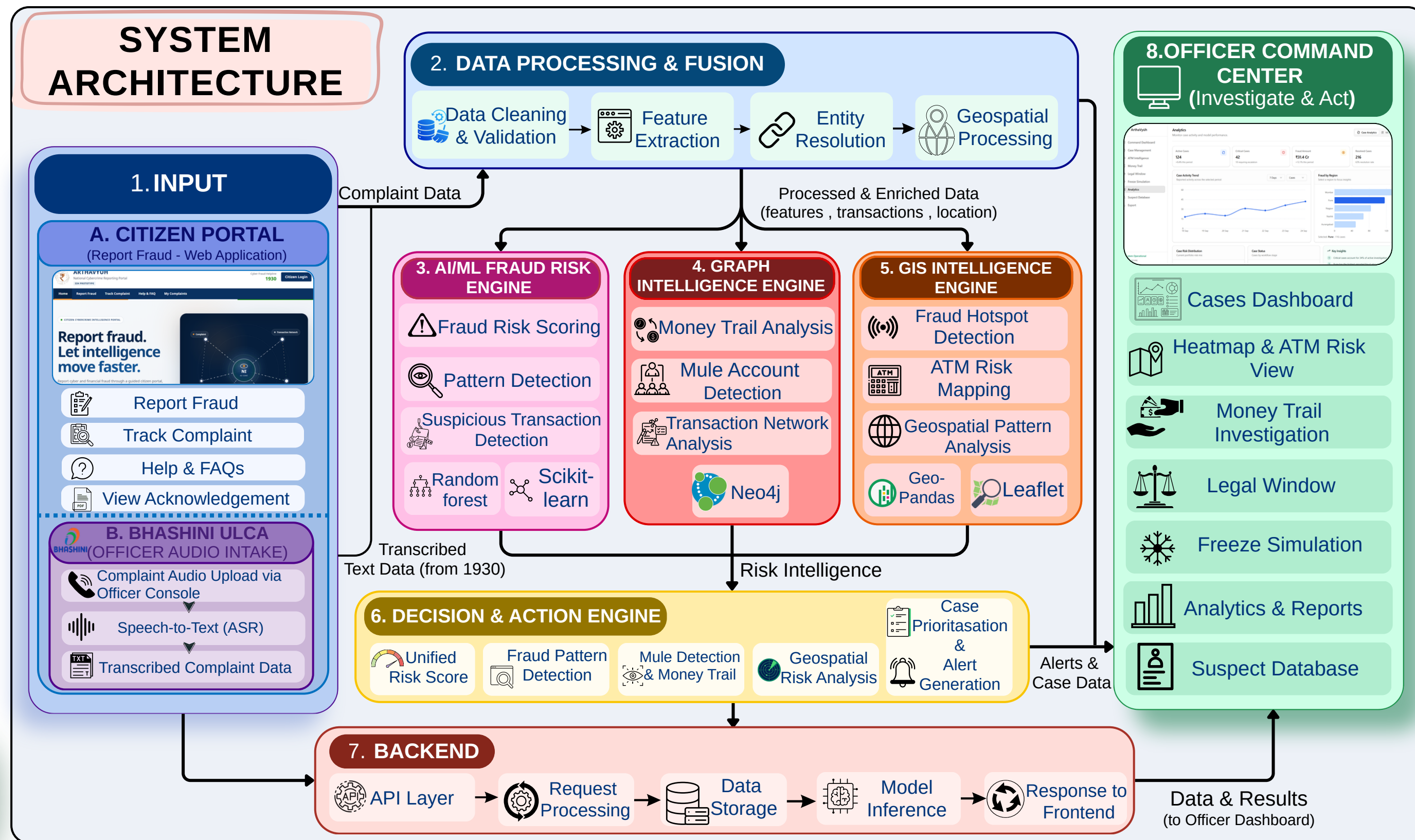
<https://youtu.be/kaSQRhFFf00?si=zawvu5UliQpFBI-M>

TECHNOLOGY STACK

- AI/ML:** Python, Scikit-learn, Pandas, NumPy
- Graph Intelligence:** Neo4j
- GIS & Mapping:** Leaflet.js, OpenStreetMap, Leaflet.heat, GeoPandas.
- Backend & APIs:** FastAPI, Node.js, Express.js, REST APIs
- Voice & Multilingual AI:** Bhashini API, Faster-Whisper
- Database:** PostgreSQL, Neo4j
- Security:** JWT, bcrypt.js
- Frontend:** React.js, Vite, Tailwind CSS, Recharts

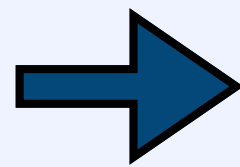
Github Link :

<https://github.com/shlok-ac/nuevion>



RISK ASSESSMENT AND MITIGATION

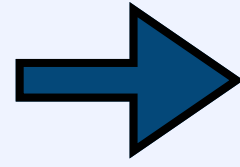
WRONGFUL
FREEZE



Use risk scoring, human verification, and configurable alert thresholds before freezing an account.



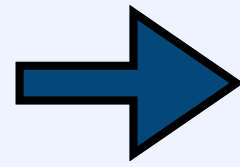
INSUFFICIENT
TRAINING DATA



Start with synthetic data for the prototype and progressively retrain/validate the model using authorized historical cases.



DATA PRIVACY
& SECURITY



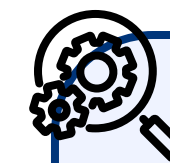
Data minimisation and role-based access to collect only necessary data and restrict sensitive information to authorized users.



RESOURCES REQUIRED

- Secure access to **privacy-protected bank transaction and ATM telemetry data**
- **Bhashini/voice-processing APIs** and **1930/NCRP** call integration
- **Coordination among banks, police, and cybercrime cells** for testing and deployment

Synthetic cash-out heatmap, Delhi, 2026.



TECHNICAL FEASIBILITY

- **Multilingual Voice Processing:** Prototype tested voice-to-JSON conversion across **5 languages: Hindi, Marathi, English, Punjabi, and Bengali.**
- **Graph-based investigation:** Prototype builds case-specific transaction graphs across 500 Synthetic Cases generated using Python Faker to trace money movement and identify terminal accounts.
- **Predictive Analytics:** Prototype uses **Random Forest ML with ATM, transaction, and mobility** features to predict likely cash-out locations.

ECONOMIC VIABILITY

- **Software-first deployment:** Designed to work with existing banking, police, GIS, and communication infrastructure, so no dedicated hardware is required.
- **Low-Cost Open Prototype:** Uses Neo4j Community, Python, and **MeitY's Bhashini to eliminate ~₹50L+ in software licensing, running at <₹1.00** compute cost per complaint.
- **Scalability:** Single-instance prototype. Planned: cyber cell → state → multi-state, with per-state nodes and queued intake.



NATIONAL CONTEXT AND & OPERATIONAL GAP

Context: CFCFRMS has saved ₹11,158 Cr across 32.80L complaints; 32.08L Layer-1 mule accounts shared with banks and FIs, and ₹25,698 Cr in transactions declined via the Suspect Registry (Source: MHA Lok Sabha reply, data to 30 Jun 2026).

The Gap: CFCFRMS primarily acts on first-hop accounts; ArthaVyūh targets layered funds (Hop-2+) and models the downstream physical ATM cash-out horizon before extraction occurs.



IMPACTS

- **Economic:** Forecasts cash-out ~45 min ahead so holds reach funds still in the banking rail.
- **Operational:** Unifies complaint, trace, forecast and approval in one dashboard, cutting manual hand-offs.
- **Investigative:** Neo4j traversal exposes layered mule accounts beyond Hop-1.
- **Social:** Aims to cut victim losses and rebuild trust in digital payments.
- **Governance:** Human-in-the-loop by design: AI advises, officers approve, actions are logged.
- **Technological:** Random Forest ranker retrain on verified outcomes as cases close.

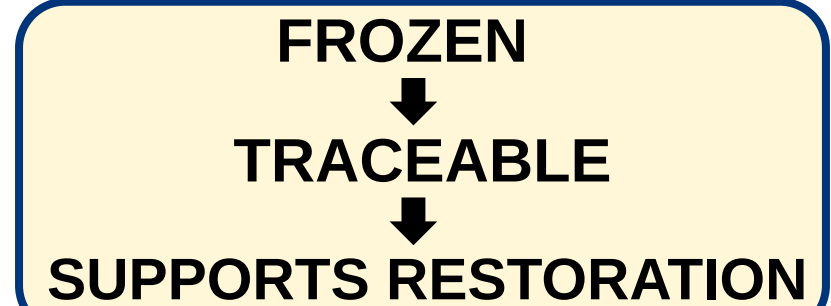
BENEFITS

Target Audience & Respective Technical Benefit

- **Police & Field Units:** Ranked top-3 ATM alerts with a modelled time window, so patrols go where cash withdrawal is most likely.
- **Banks & FIs:** One officer-approved, drafted lien request instead of scattered calls.
- **Cyber Investigators:** Drafted §106 BNSS notice for legal review, so less time goes on paperwork.
- **Victims & Citizens:** Report by form or Bhashini voice; approved hold aims to secure funds early.

FUTURE PROSPECTS

- **Real-Data Pilot:** Validate accuracy on anonymised **NCRP records** starting within a single state cyber cell.
- **Live Bank Integration:** Upgrade freeze simulations to operational debit-liens via official **CFCFRMS** rails.
- **Continuous Learning:** Automatically retrain the ATM predictor on verified closed cases to sharpen accuracy over time.
- **National Topology Scale:** Expand interstate graph linkage so a single complaint exposes connected mule networks nationwide.



Reactive → Predictive. Isolated → Connected. Hours → Minutes.

1. EXISTING ECOSYSTEM (Government & Financial Systems)

Existing systems already address important parts of cyber-financial fraud.


CFCFRMS / 1930
I4C – Ministry of Home Affairs
Financial-fraud reporting, fund tracing and lien coordination with banks.

 **ArthaVyūh adds →**
Physical Cash-out Forecast

 [Official Portal](#) ↗


I4C Suspect Registry
Shares suspect identifiers and Layer-1 mule-account intelligence with participating banks and financial institutions

 **ArthaVyūh adds →**
Case-specific ATM Forecast

 [Check Suspect](#) ↗


RBIH MuleHunter.ai™
Supervised ML solution for near-real-time identification of mule bank accounts.

 **ArthaVyūh adds →**
Post-mule Cash-out Prediction

 [RBI Annual Report](#) ↗

2. RESEARCH EVIDENCE (What we BORROW → What we ADD)

	Demiriz & Ekizoğlu (2016) Fuzzy rule-based analysis of spatio-temporal ATM usage data for fraud detection and prevention. <i>Journal of Intelligent & Fuzzy Systems</i> , 31(2), 805–813 DOI: 10.3233/JIFS-169012	Borrow : ATM usage patterns over space and time. Add : Fraud-case-conditioned ranking of likely cash-out ATMs.	 Read Paper ↗
	Abdul Rani et al. (2025) Dissecting the post-investigation actions on money mule cases in banking financial crime compliance. <i>Journal of Money Laundering Control</i> , 28(6), 747–763 DOI: 10.1108/JMLC-06-2025-0105	Borrow : Post-investigation mule handling. Add : Spatial prioritization of the likely cash-out point.	 Read Paper ↗
	Butt et al. (2020) Spatio-Temporal Crime HotSpot Detection and Prediction: A Systematic Literature Review. <i>IEEE Access</i> , 8, 166553–166574 DOI: 10.1109/ACCESS.2020.3022808	Borrow : Spatio-temporal hotspot prediction principles. Add : Case-level (not area-level) prioritisation.	 Read Paper ↗
	Europol IOCTA 2026 Threat Context Cybercrime and online/payment-fraud threat landscape, covering emerging technologies and evolving threats. <i>Europol EC3</i> • 28 Apr 2026	Provides Threat Context on Cybercrime, Payment fraud and emerging technologies	 Read Report ↗

3. IDENTIFIED GAP → ArthaVyūh CONTRIBUTION

THE MISSING OPERATIONAL STEP
Traced exit account → Physical cash-out location

